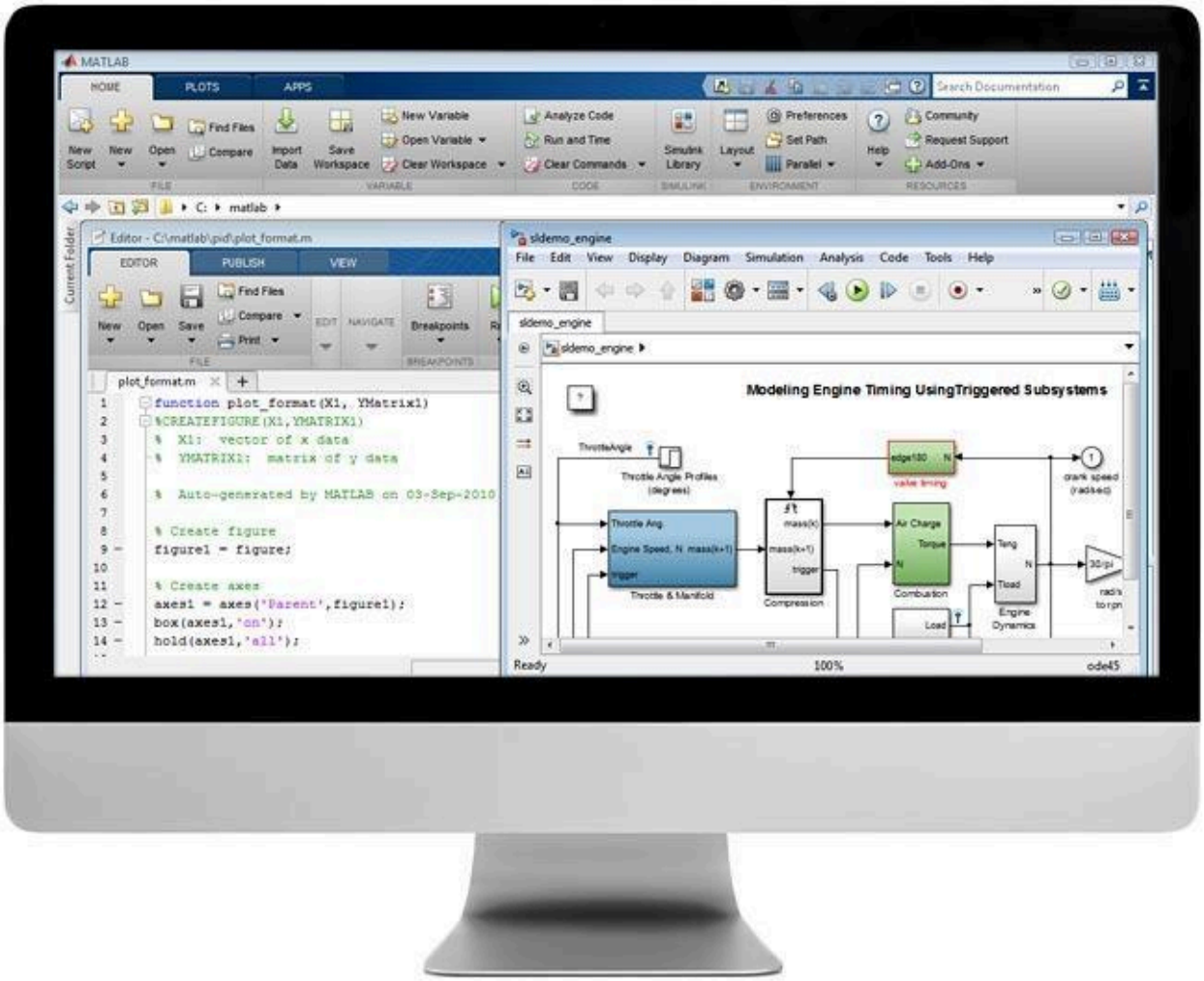
MATLAB Simulation & Application Software Development: Beginner-to-Job Roadmap

Your target: AutoMech Solutions / EV / Automotive Embedded Engineering Your current level: Beginner — no MATLAB experience Recommended learning duration: 12 weeks, studying 2–3 hours daily Budget: Start with free learning resources and free software access where available.

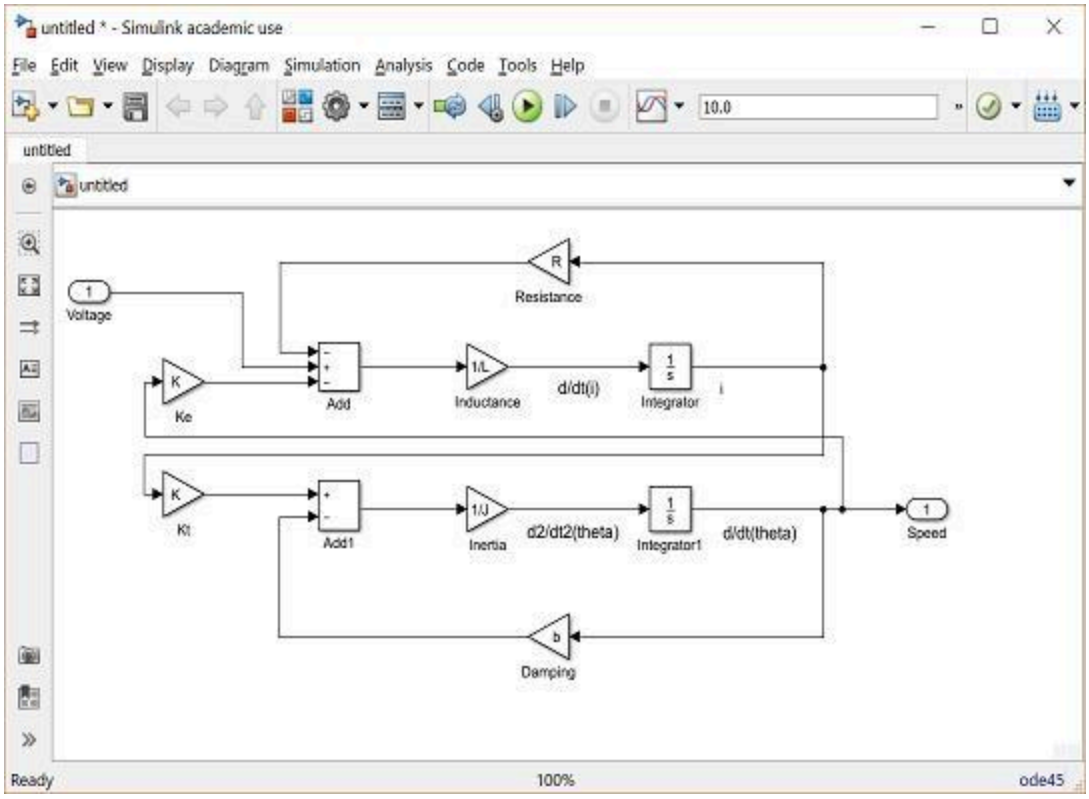
Your goal should be to learn MATLAB from scratch, understand how electrical and automotive systems work, simulate those systems, and eventually develop small engineering applications. This will help you prepare for roles involving EV chargers, power electronics, embedded hardware testing, BMS, and automotive control systems.

1. What should you learn?



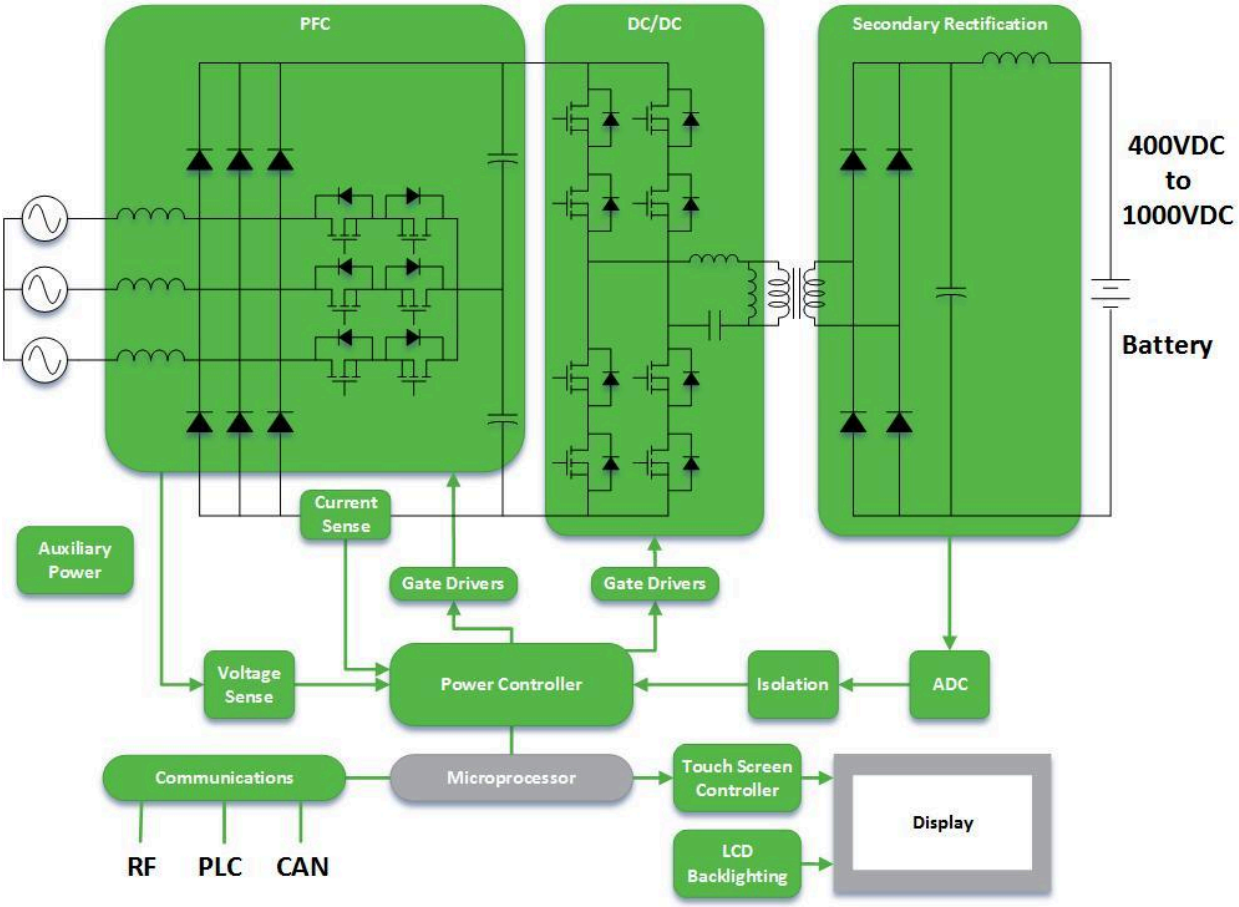
1. MATLAB Programming

Learn variables, arrays, matrices, conditions, loops, functions, and plotting. These are the foundations of engineering calculations and simulation.



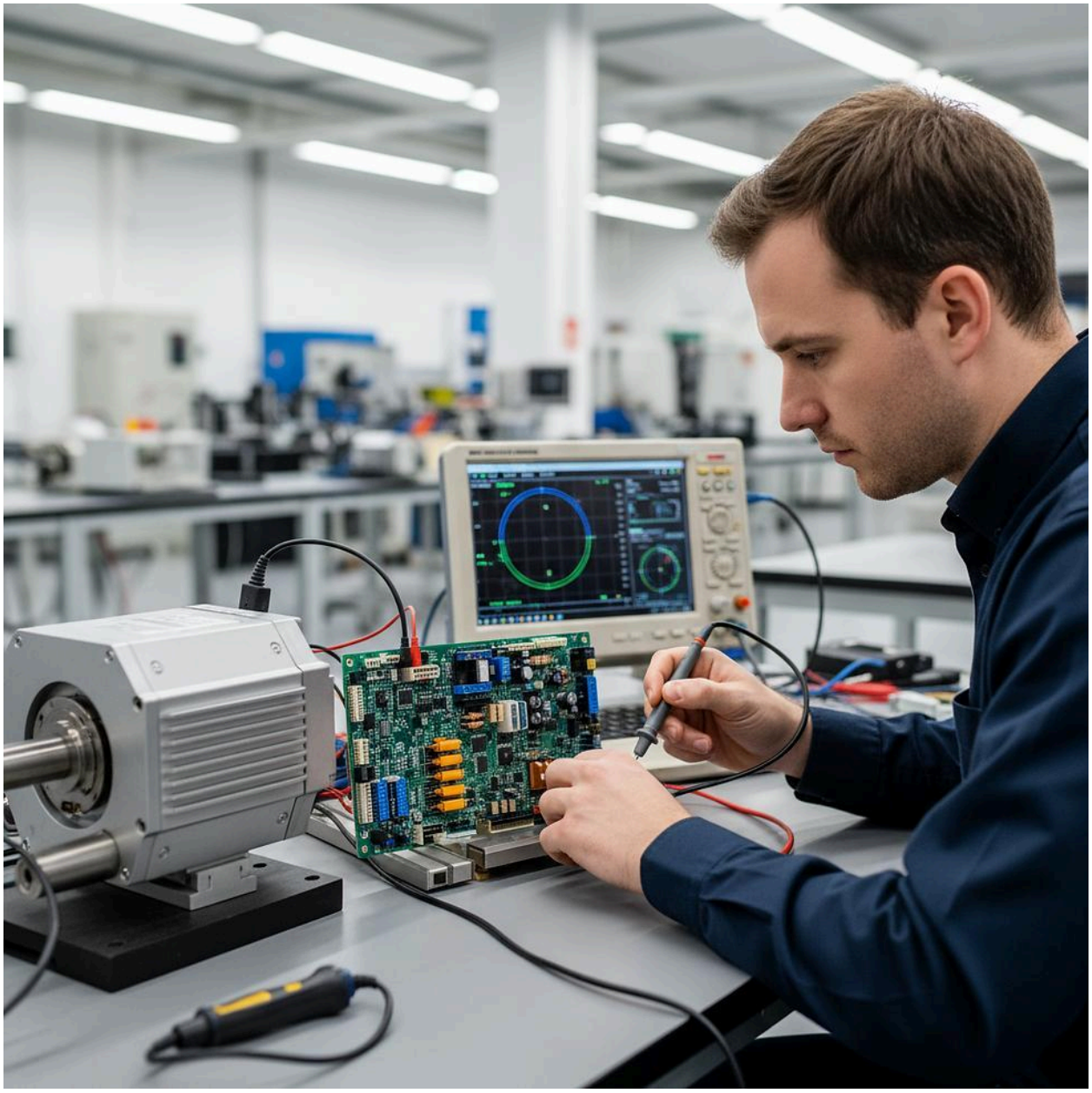
2. MATLAB Simulation and Simulink

Learn to create block diagrams and simulate a motor, battery, control system, or electrical circuit without immediately building physical hardware.



3. Electrical and EV Systems

Understand voltage, current, power, batteries, DC-DC buck/boost converters, charging systems, sensors, and feedback control.



4. Application Software Development

Write programs that accept inputs, perform engineering calculations, display graphs, and help engineers test or monitor a system.

2. Your 12-week learning plan

Weeks 1–2

Phase 1 — MATLAB Fundamentals

- Learn the MATLAB interface and Command Window.
- Variables, arrays, matrices, arithmetic, and units.
- `if`, `for`, `while`, and functions.
- Create plots and write simple scripts.

Practice: Calculate voltage, current, power, resistance, and battery energy.

Weeks 3–4

Phase 2 — Engineering Calculations and Data Analysis

- Plot voltage, current, temperature, and time.
- Read data from CSV files.
- Calculate averages, maximums, and energy consumption.
- Compare measured and simulated results.

Practice: Analyze EV battery voltage and current data and generate graphs.

Weeks 5–6

Phase 3 — Simulink Fundamentals

- Understand blocks, signals, and connections.
- Build a block diagram.
- Run a simulation and adjust parameters.
- Observe system behaviour using Scope and plots.

Practice: Simulate a simple DC motor or battery response.

Weeks 7–8

Phase 4 — EV and Power Electronics

- DC-DC buck and boost converter principles.
- Battery state of charge (SOC) basics.
- Voltage regulation and feedback control.
- Sensors, operating limits, and fault conditions.

Practice: Model a basic battery charging system and study how its voltage responds to changes in load.

Weeks 9–10

Phase 5 — Application Software Development

- Organize code into reusable functions.
- Build a simple graphical interface.

- Accept user inputs and display calculated results.
- Validate inputs and handle errors.

Practice: Create an EV battery calculator or charger monitoring dashboard.

Weeks 11–12

Phase 6 — Job-Focused Project and Portfolio

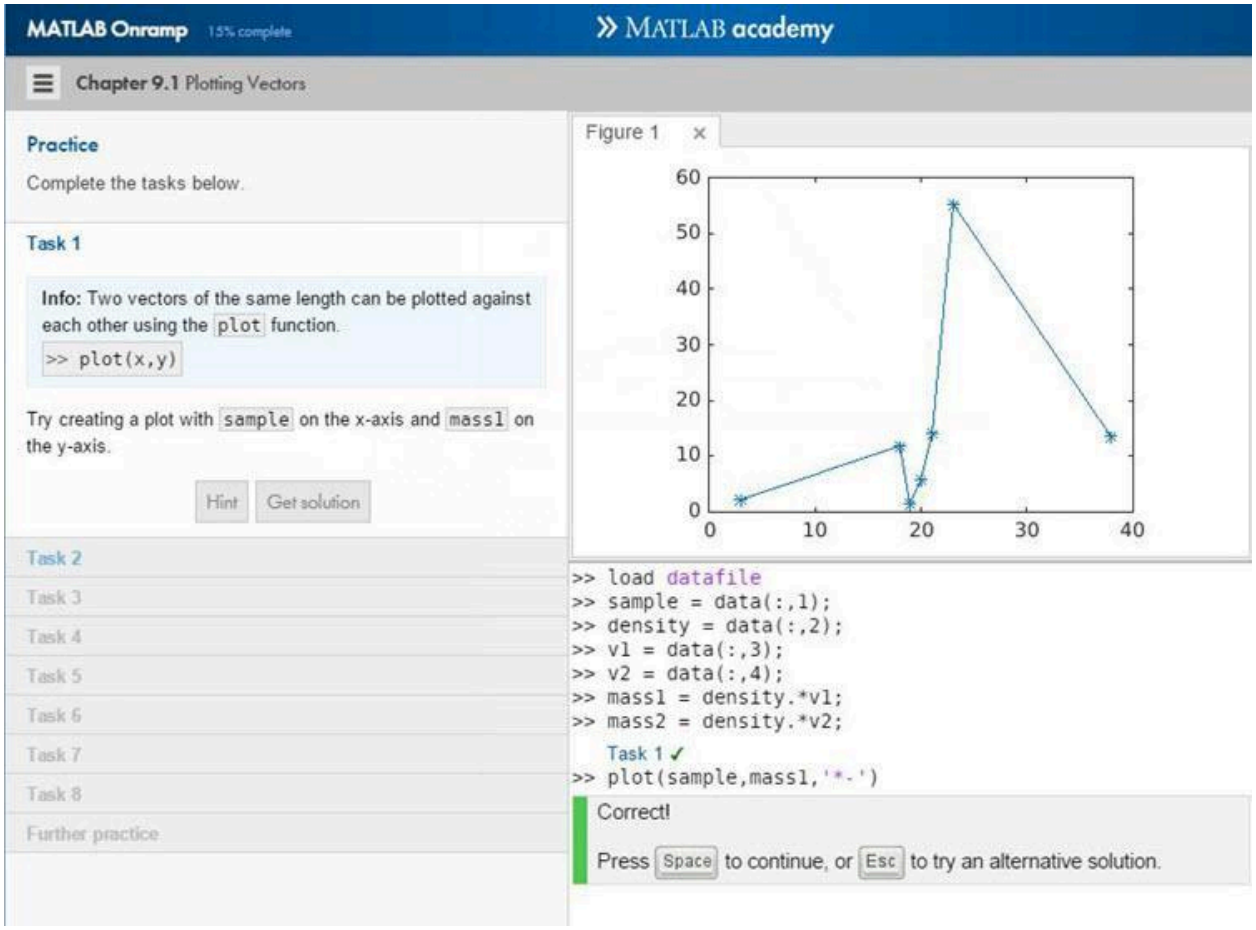
- Complete one end-to-end engineering project.
- Test different operating conditions and fault cases.
- Document the model, assumptions, results, and limitations.
- Upload your code, graphs, and project report to GitHub.

Practice: Build a simulated EV charger monitoring and protection system.

This is a practical learning schedule, not a guarantee of job readiness in 12 weeks. If you are new to both MATLAB and power electronics, allow extra time for the EV simulation phases.

3. Free learning resources

Start with official MathWorks courses before paying for any training.



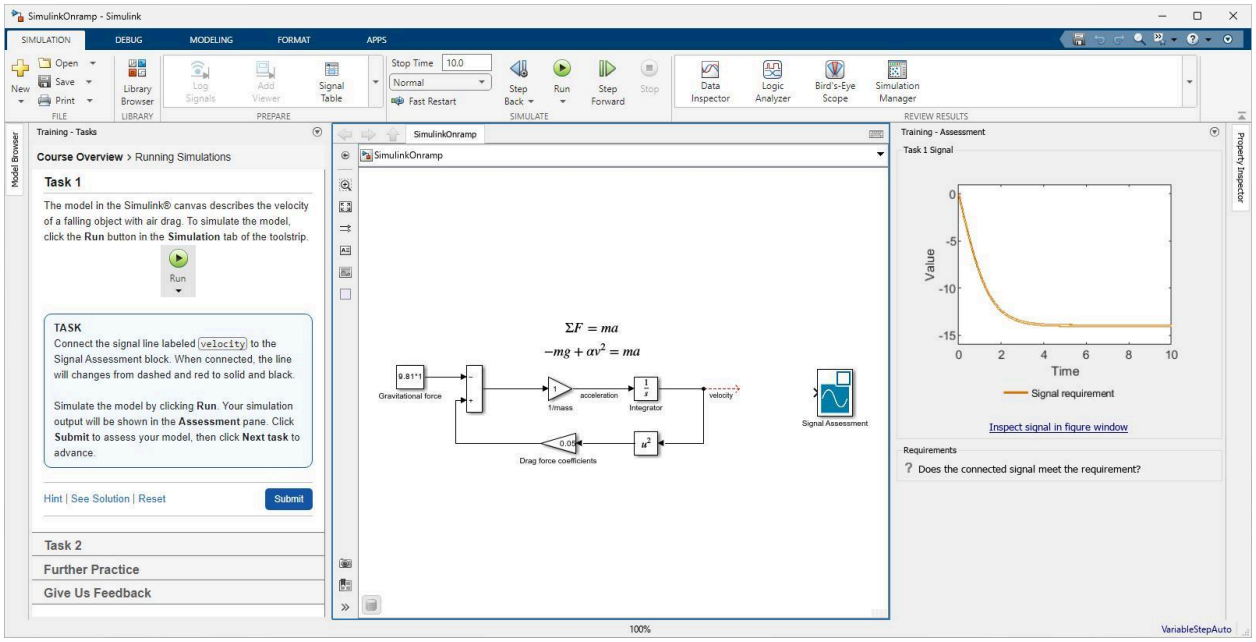
1. MATLAB Onramp

Start here

A beginner-friendly, interactive introduction to MATLAB. The course is free with a MathWorks account.

MATLAB & Simulink
+1

[Open MATLAB Onramp]

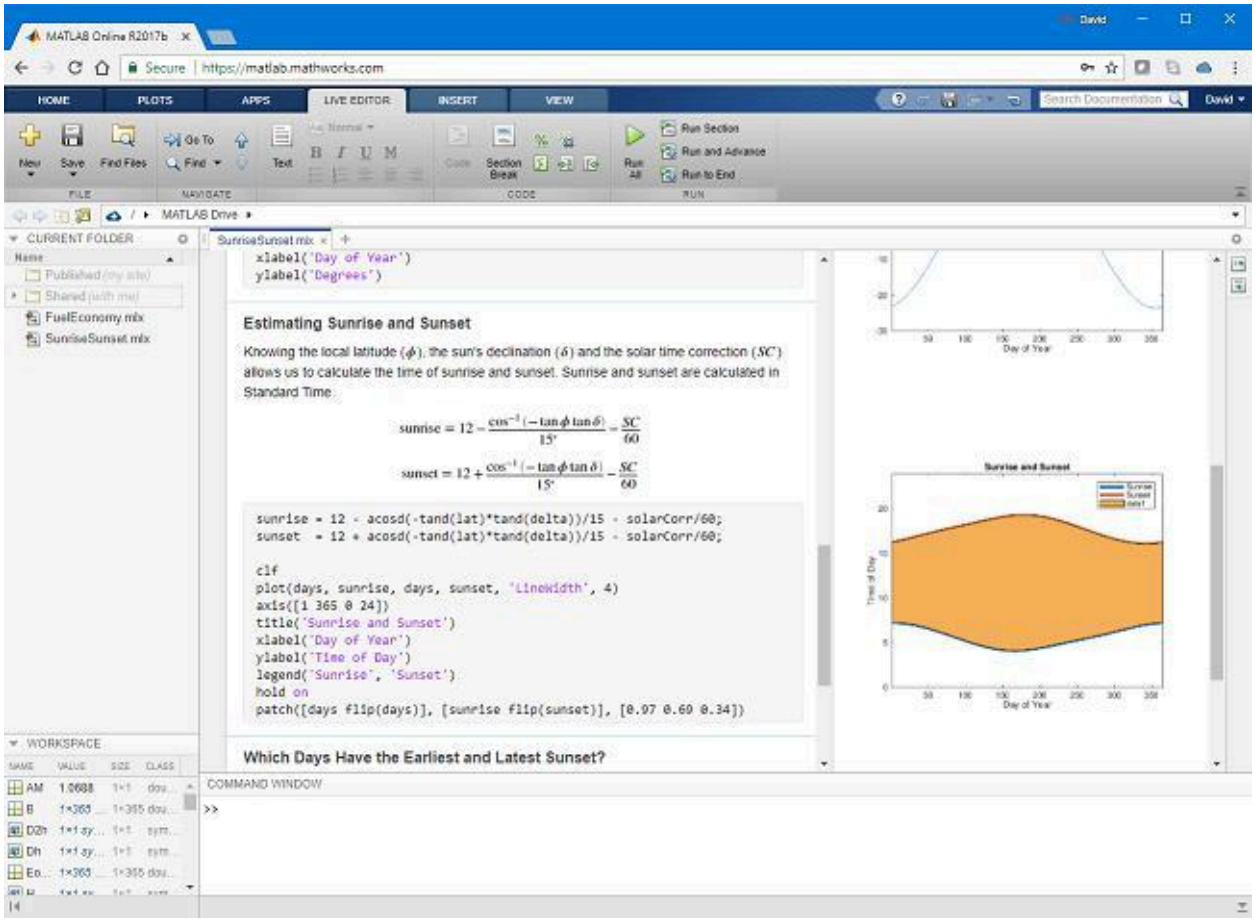


2. Simulink Onramp

Learn block diagrams, signals, and basic dynamic-system simulation. The introductory Onramp is free.

MATLAB
+1

Open Simulink Onramp

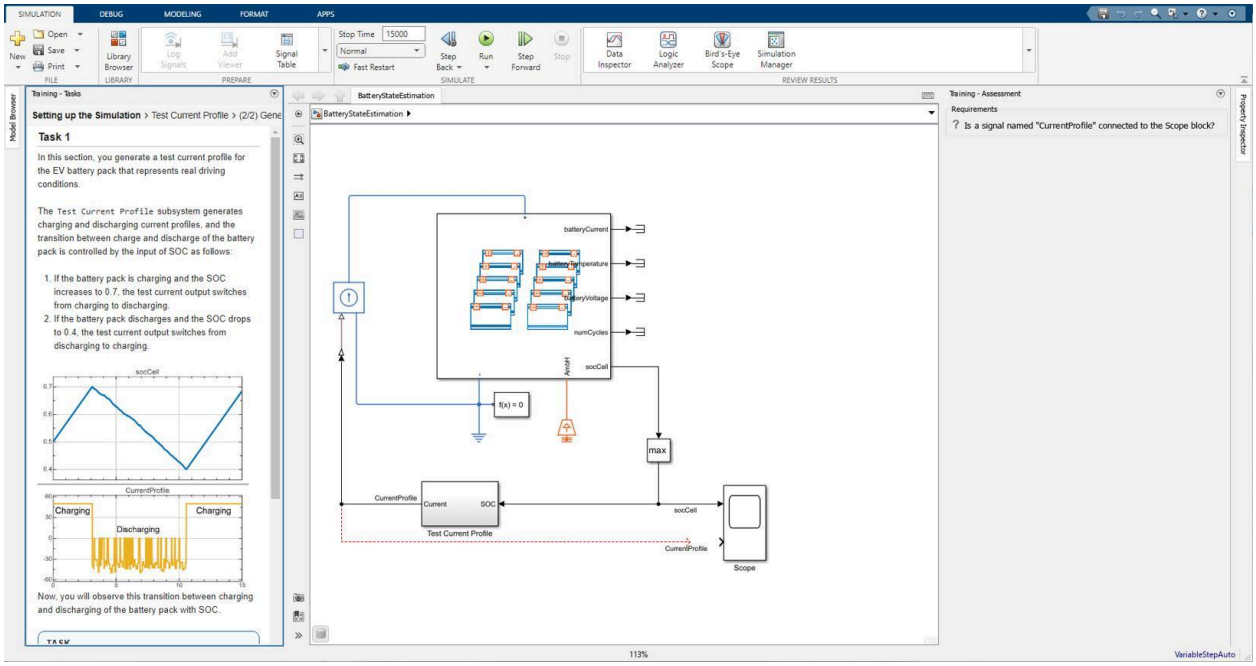


3. MATLAB Online

Run MATLAB in your browser without installing desktop software. The Basic version currently offers 20 hours per month of free use; product availability is limited.

MATLAB & Simulink
+1

[Open MATLAB Online]



4. Automotive and EV Simulation Examples

Explore how MATLAB and Simulink are used to model batteries, vehicle electrical systems, motors, and control algorithms. Some examples require additional licensed toolboxes.

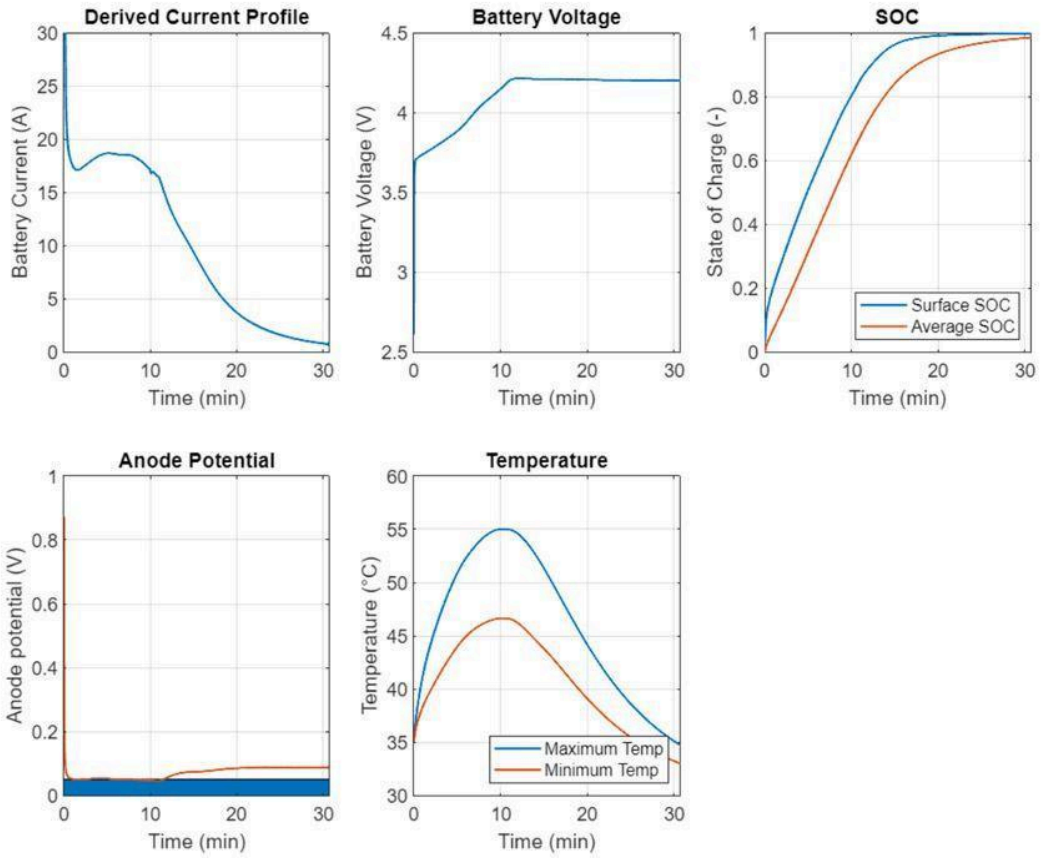
MATLAB & Simulink
+2

Explore automotive applications

Important: Check whether your MCA college provides a MATLAB/Simulink campus licence. That may give you access to tools that are not included in the free MATLAB Online Basic version.

4. Five practical projects for your portfolio

Build these in order, starting with simple MATLAB scripts and gradually moving to Simulink.

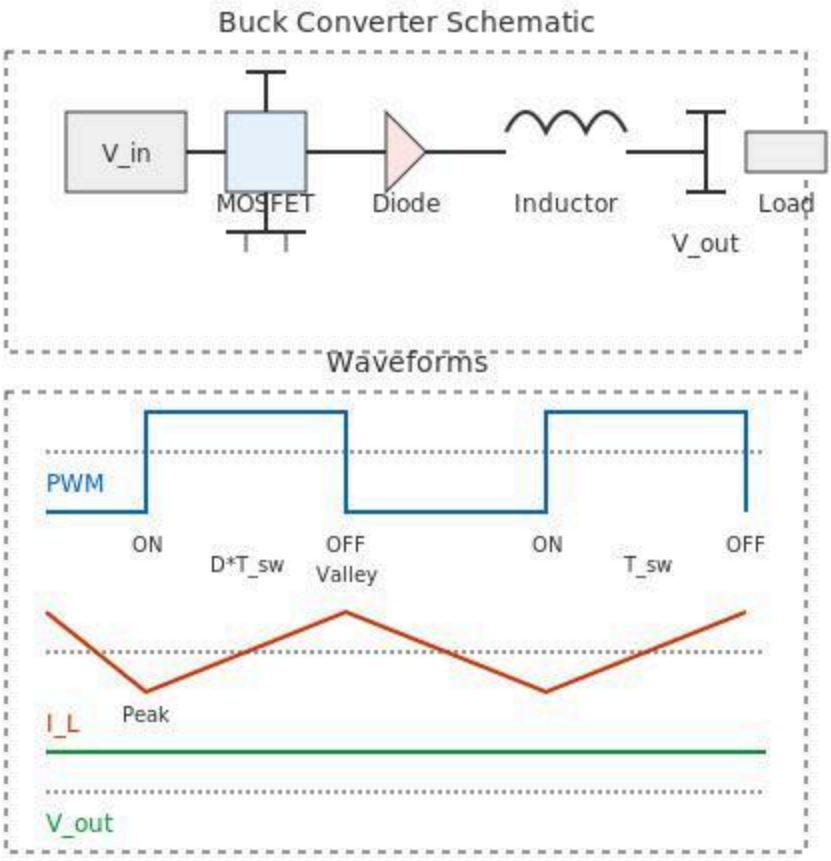


Project 1 — EV Battery Performance Calculator

Beginner

- Enter battery voltage, capacity, and load current.
- Calculate power, approximate runtime, and energy.
- Plot battery voltage or SOC against time using assumed data.

Skills: MATLAB variables, formulas, functions, and plots.



Project 2 — Buck and Boost Converter Analysis

Beginner–Intermediate

- Calculate ideal output voltage at different duty cycles.
- Plot input versus output voltage.
- Compare ideal calculations with a Simulink model.

Skills: Power electronics, equations, simulation, and result analysis.



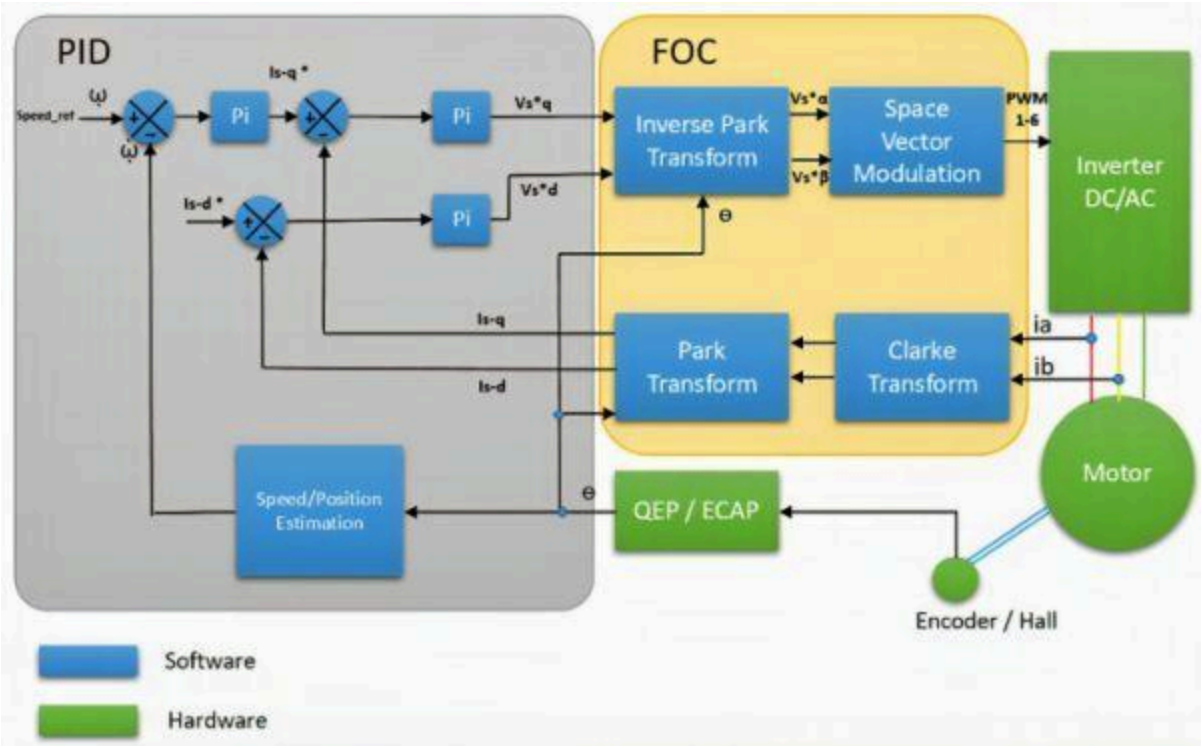
Project 3 — EV Charger Monitoring and Fault Detection

Job-focused

- Simulate charger voltage and current measurements.
- Flag overvoltage, overcurrent, and overtemperature conditions.
- Display normal, warning, and fault states.

Skills: MATLAB logic, Simulink, testing, and engineering documentation.

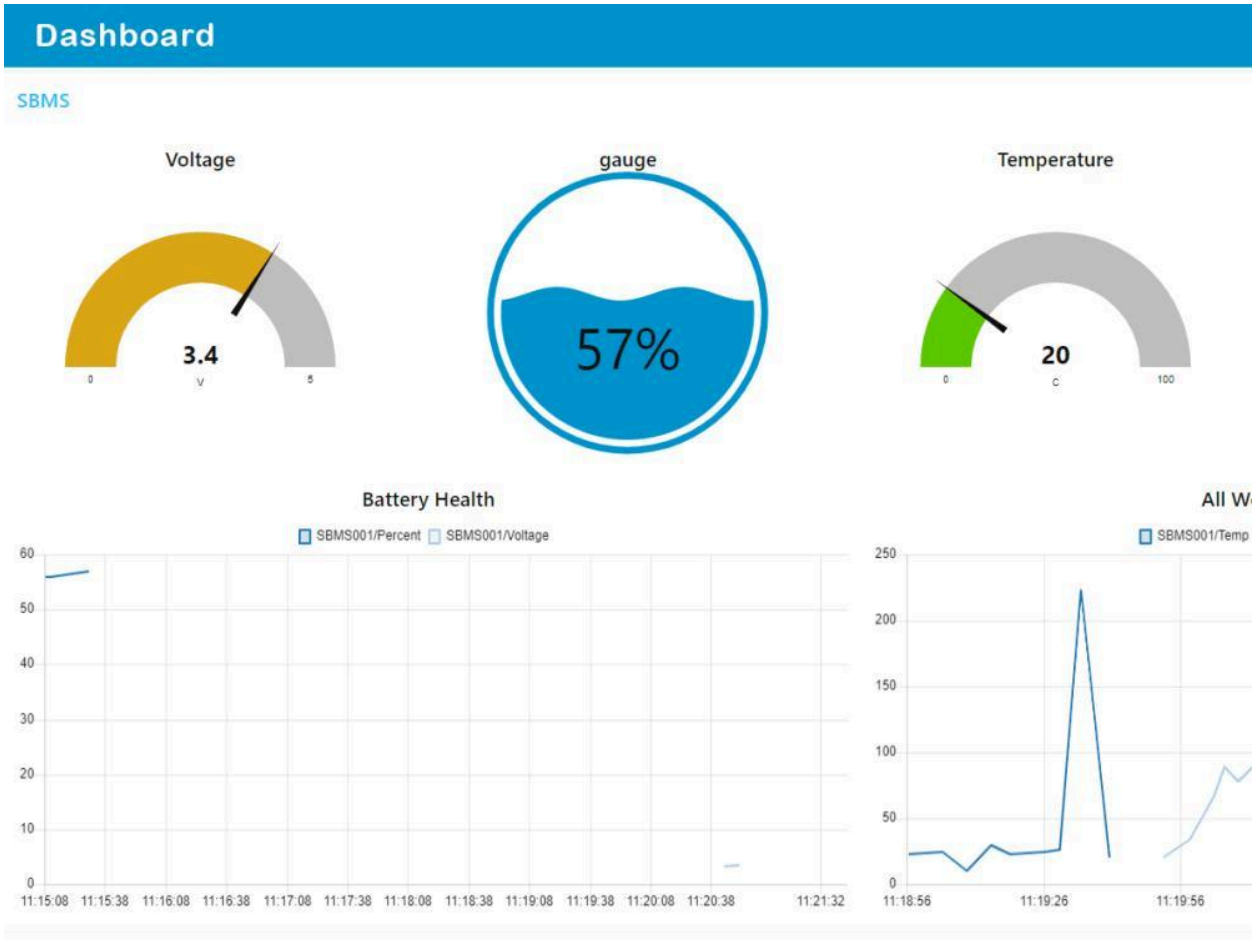
This is a simulation project, not a certified charger protection design.



Project 4 — DC Motor Speed Control

- Model a simple motor and its response to input changes.
- Compare open-loop and feedback control.
- Study how controller settings affect speed and settling time.

Skills: Control systems, Simulink, and automotive embedded fundamentals.



Project 5 — EV Battery Testing Application

- Build a basic graphical interface.
- Import test data from a CSV file.
- Display voltage, current, temperature, and warning messages.
- Export a summary report.

Skills: Application development, data processing, plotting, and testing.

5. What does Application Software Development mean?

For your target roles, it can mean writing software that helps engineers calculate, simulate, test, or monitor automotive and EV systems.

For example, your application could work like this:

Input

Battery voltage, current, temperature, and limits

MATLAB calculations

Calculate power, energy, and operating status

Application interface

Display graphs, measurements, warnings, and results

Learn these technologies in this order:

1. MATLAB scripts and functions — the foundation.
2. Simulink — model and simulate dynamic systems.
3. MATLAB App Designer — create graphical applications, subject to licence availability.
4. Python and C/C++ — useful for broader application development and embedded systems.
5. Hardware integration — later, learn how to work with sensors, microcontrollers, CAN data, and real measurements.

MATLAB is not a replacement for every application development technology. Your goal is to use it where it is useful for engineering calculations, simulation, analysis, and testing.

6. Daily study routine

Aim for 2 hours daily, six days per week.

Time	Activity
30 minutes	Watch lessons or study concepts
45 minutes	Write code or build a simulation
30 minutes	Complete an engineering exercise
15 minutes	Document what you learned

Every week, save your code and write a short note explaining what you built, what worked, and what you need to improve.

7. How this connects to AutoMech and EV jobs

Job area	Most relevant skills
MATLAB/Simulation Intern	MATLAB, Simulink, plotting, model validation
EV/Power Electronics Intern	Circuit fundamentals, converters, simulation
Automotive Embedded Intern	C/C++, sensors, control logic, basic CAN
Test and Validation Intern	Test cases, data analysis, fault detection, reports
Application Software Intern	Programming, interfaces, data processing, debugging

For the kind of embedded hardware and EV charger work you are targeting, prioritize MATLAB + Simulink + power electronics + testing fundamentals. Add PCB design and practical hardware debugging separately if the role involves board development.

Learning MATLAB alone will not qualify you for every role, but a portfolio of relevant projects can demonstrate your progress and give you practical examples to discuss in interviews.

8. Your first assignment — start today

You do not need to understand all of MATLAB before beginning.

1. Open MATLAB Online.
2. Create or sign in to your MathWorks account.
3. Start MATLAB Onramp.
4. Learn basic variables and arithmetic.
5. Try this small EV engineering calculation:

```
% Simple EV electrical calculations

voltage = 400;       % Volts
current = 20;        % Amperes
capacity = 60;       % Battery capacity in kWh
power = voltage * current / 1000;

fprintf('Power: %.1f kW\n', power);

runtime = capacity / power;
fprintf('Ideal runtime: %.2f hours\n', runtime);
```

Expected output:

```
Power: 8.0 kW
Ideal runtime: 7.50 hours
```

This runtime is an ideal estimate assuming constant power and the full stated battery capacity is usable. Real-world runtime will differ because of losses, usable capacity, and changing loads.

Your first milestone: Finish MATLAB Onramp, write five small engineering programs, and then move to Simulink.